## 12.5 Probability Models and Expected Value

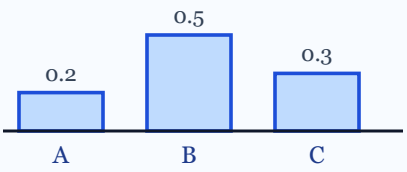
A **probability model** assigns a probability to every outcome in a sample space; it's valid only if every probability is between $0$ and $1$ and they all sum to exactly $1$. **Theoretical probability** comes from a formula or equally-likely-outcomes assumption; **experimental probability** comes from actual trial data ($\frac{\text{occurrences}}{\text{trials}}$) — the Law of Large Numbers says the experimental value tends to approach the theoretical one as trials increase. **Expected value**, $E = \sum(\text{outcome} \times \text{its probability})$, is the long-run average result and is central to deciding whether a game, raffle, or insurance decision favors you or the house — even though it may never equal any single actual outcome.

# Examples

### Example 1: Checking a valid model

Outcomes $A, B, C$ have probabilities $0.2, 0.5, 0.3$. Is this a valid probability model?

**Solution**




Each bar is between 0 and 1, and together they account for the whole sample space.

Check two conditions for a valid model: (1) each probability is between $0$ and $1$ — yes for all three; (2) they sum to exactly $1$: $0.2 + 0.5 + 0.3 = 1$. ✓

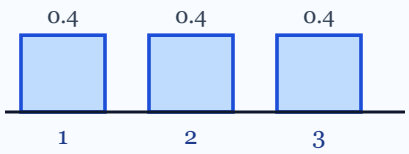
**Answer:** Yes, valid.

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## Example 2: Spotting an invalid model

Probabilities $0.4$, $0.4$, $0.4$ are proposed for three outcomes. Valid?

### Solution




Each individual bar looks fine (between 0 and 1), but the required check is whether they sum to exactly 1.

Each value is between $0$ and $1$ on its own, but check the sum: $0.4 + 0.4 + 0.4 = 1.2 \neq 1$.

**Answer:** No — the probabilities must sum to exactly $1$, and $1.2 \neq 1$.

## Example 3: Experimental probability

A coin is flipped $80$ times, landing heads $36$ times. Find the experimental $P(\text{heads})$.

### Solution

$$P_{\text{exp}}(\text{heads}) = \frac{36}{80} = \frac{9}{20} = 0.45.$$

**Answer:** $0.45$ (or $\frac{9}{20}$).

## Example 4: Theoretical probability, for comparison

For the same fair coin, what is the theoretical $P(\text{heads})$?

### Solution

A fair coin has two equally likely outcomes: $P(\text{heads}) = \frac{1}{2} = 0.5$.

**Answer:** $0.5$. *Compare to Example 3's $0.45$ — with only $80$ flips, experimental results can still drift from the theoretical value.*

### Example 5: Expected value — two outcomes

A game pays \$10 with probability $0.3$, and \$0 otherwise. Find the expected value.

**Solution**

Identify the formula: $E = \sum(\text{outcome} \times \text{its probability})$, summed over every possible outcome.

Compute: $E = 10(0.3) + 0(0.7) = 3 + 0 = 3$.

**Answer:** \$3 — the long-run average payout per play.

### Example 6: Is the game fair?

A game costs \$4 to play and has expected payout \$4. Is it fair?

**Solution**

A game is **fair** exactly when the expected payout equals the cost to play (expected net gain $= 0$).

$4 = 4$, so yes.

**Answer:** Fair.

### Example 7: Expected net gain (or loss)

A game costs \$5 to play; it pays \$20 with probability $0.2$, and \$0 otherwise. Find the expected net gain.

**Solution**

**Step 1 — expected payout:** $E(\text{payout}) = 20(0.2) + 0(0.8) = 4$.

**Step 2 — expected net:** subtract the cost: $4 - 5 = -1$.

**Answer:** expected net $= -1$ (an average loss of \$1 per play — the game favors the house).

### Example 8: Law of Large Numbers

As the number of trials increases, what happens to the experimental probability?

**Solution**

It tends to get closer and closer to the theoretical probability — large samples smooth out the random "noise" that small samples show (like Example 3's $0.45$ vs. the true $0.5$).

**Answer:** it approaches the theoretical probability.

## Example 9: Finding a missing probability

A spinner model has $P(\text{red}) = \frac{1}{2}$, $P(\text{blue}) = \frac{1}{3}$, and $P(\text{green}) = x$. Find $x$.

**Solution**

All probabilities must sum to $1$:

$$\frac{1}{2} + \frac{1}{3} + x = 1$$

Common denominator $6$: $\frac{3}{6} + \frac{2}{6} + x = 1 \Rightarrow \frac{5}{6} + x = 1 \Rightarrow x = \frac{1}{6}$.

**Answer:** $x = \frac{1}{6}$.

## Example 10: Multi-tier expected value — a raffle

1000 raffle tickets are sold. One grand prize of \$500 is drawn, and 5 second prizes of \$50 each are also drawn (all from the same 1000 tickets, no ticket wins twice). Each ticket costs \$1. Find the expected winnings per ticket, and whether buying a ticket is a good deal on average.

### Solution

| Outcome | Probability | Payout |
|---|---|---|
| **Grand prize** | 1/1000 | 500</td></tr><tr><td style="border:1px solid #94a3b8;padding:4px 12px;text-align:center;font-weight:700;background:#f8fafc">Second prize (5 of them)</td><td style="border:1px solid #94a3b8;padding:4px 12px;text-align:center;">5/1000</td><td style="border:1px solid #94a3b8;padding:4px 12px;text-align:center;">50 each |
| **No prize** | 994/1000 | 0</td></tr></table> Expected value sums each outcome's probability times its payout across every row. Grand prize term: 500\times\dfrac{1}{1000}=0.50. < *br* >< *b* > *Secondprizeterm* :< /*b* >50\times\dfrac{5}{1000}=50\times\dfrac{1}{200}=0.25.<br>All other tickets win \0, contributing nothing to the sum. $E(\text{winnings}) = 0.50 + 0.25 = 0.75$, i.e. \$0.75. Expected net $= 0.75 - 1 = -0.25$, i.e. a loss of \$0.25 per ticket. **Answer:** $E = 0.75$ dollars; not a good deal on average — every ticket loses \$0.25 in expectation (typical for raffles, since the organizer keeps a share). |

**Example 11: Expected value in a real decision — is a warranty worth it?**

A $500 laptop has an 8% chance of needing a $300 repair within a year. An extended warranty covering that repair costs $35. Using expected value, decide whether buying the warranty is the better choice on average.

**Solution**

**Expected repair cost without the warranty:** $E = 300(0.08) + 0(0.92) = 24$.

**Cost with the warranty:** a guaranteed $35 (assuming it fully covers the repair).

Since the expected repair cost ($24) is less than the warranty ($35), skipping the warranty is cheaper *on average* — by $35 - 24 = 11$ dollars.

**Answer:** skipping the warranty saves $11 in expectation.

*Caveat: expected value describes the long-run average — someone risk-averse might still prefer the warranty's certainty over the small chance of a $300 hit.*

## Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

### Problem 1: Valid model?

Probabilities 0.1, 0.6, 0.3. Valid?

**Answer:** **Verification:**

### Problem 2: Valid model?

Probabilities 0.5, 0.5, 0.2. Valid?

**Answer:** **Verification:**

### Problem 3: Experimental

50 trials, event occurs 18 times. Find the experimental probability.

**Answer:** **Verification:**

### Problem 4: Theoretical

Fair die. Find the theoretical $P(6)$.

**Answer:** **Verification:**

### Problem 5: Expected value

A game pays \$8 with probability 0.25, else \$0. Find $E$.

**Answer:** **Verification:**

### Problem 6: Fair game?

Cost \$2; expected payout \$2. Fair?

**Answer:** **Verification:**

### Problem 7: Expected net

Cost \$3; pays \$12 with probability $0.2$. Find the expected net.

**Answer:** **Verification:**

### Problem 8: Find $x$

Model: $0.4$, $0.35$, $x$. Find $x$.

**Answer:** **Verification:**

### Problem 9: Experimental

200 flips, 108 heads. Find the experimental $P$(heads).

**Answer:** **Verification:**

### Problem 10: Expected value

Outcomes \$1 and \$5 with probabilities 0.8 and 0.2. Find $E$.

**Answer:** **Verification:**

### Problem 11: Fair game?

Cost \$5; expected payout \$4. Fair?

**Answer:** **Verification:**

### Problem 12: Law of Large Numbers

As trials increase, experimental probability tends toward what?

**Answer:** **Verification:**

### Problem 13: Model rule

In a valid probability model, what must all outcome probabilities sum to?

**Answer:** **Verification:**

### Problem 14: Raffle

500 tickets sold; 1 prize of $250 and 10 prizes of $10 each. Find the expected winnings per ticket.

**Answer:** **Verification:**

### Problem 15: Challenge

A $500 laptop has an 8% chance of needing a $300 repair. An extended warranty costs $35. Using expected value, determine whether the warranty is a good deal on average, and by how much.

**Answer:** **Verification:**

## ✓ Answer Key

| | | | |
|---|---|---|---|
| 1. | Yes | 9. | 0.54 |
| 2. | No (sum = 1.2) | 10. | $1.80 |
| 3. | $\frac{9}{25}$ | 11. | No — favors the house |
| 4. | $\frac{1}{6}$ | 12. | The theoretical probability |
| 5. | $2 | 13. | 1 |
| 6. | Yes | 14. | $0.70 |
| 7. | −0.60 (a loss of $0.60) | 15. | Not a good deal on average; skipping it saves $11 in expectation |
| 8. | 0.25 | | |

**Great work — keep practising!**